

Through the Lens

4 Years of Campus Life in 57k Photos

Process Book

Adrien Buttier · Martina Gatti · Lea Grieder

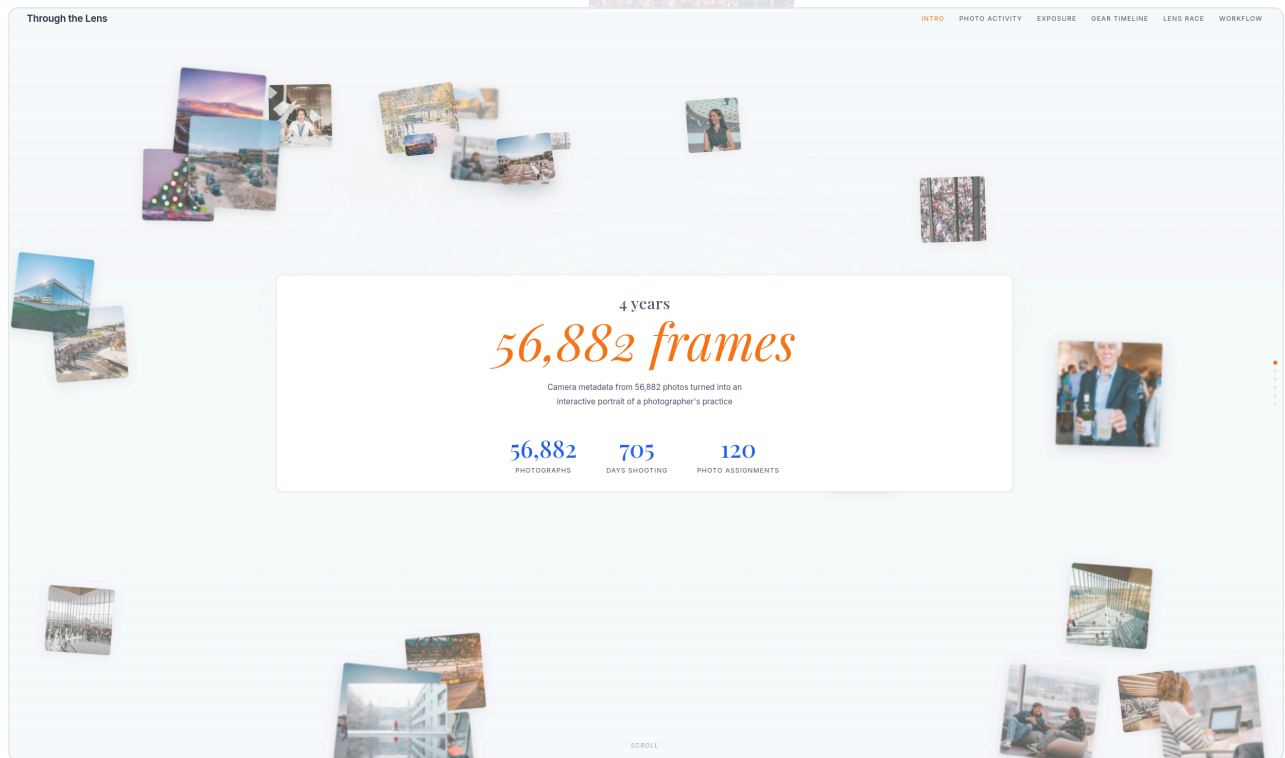
Live site: com-480-data-visualization.github.io/through_the_lens/

56,882
PHOTOGRAPHS

4
YEARS

952
METADATA FIELDS

5
VISUALIZATIONS



Where the idea came from

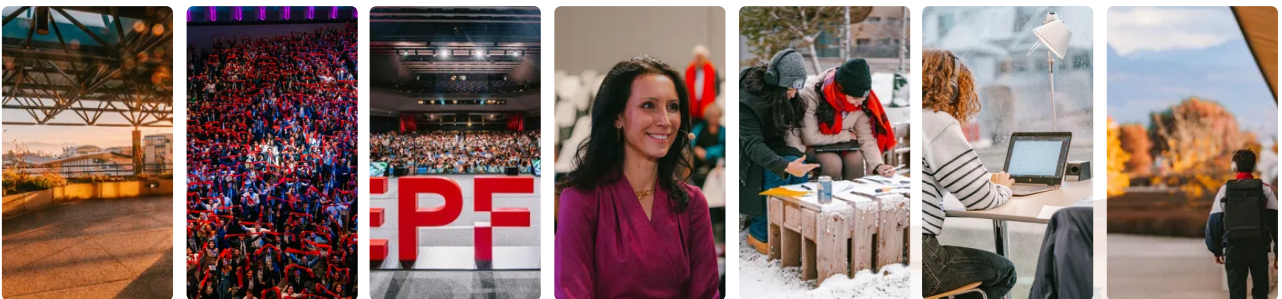
The project started with Adrien's pitch: he had spent over four years photographing EPFL campus life (events, concerts, associations, sport, studying, everyday moments) and had never thought of looking at the metadata as data. Lightroom had been recording everything silently: 56,882 frames, each stamped with ISO, aperture, shutter speed, focal length, every post-processing slider touched. He knew the photographs. He had no idea what the numbers said.

One external project shaped how we thought about showing it. The Pudding's [*Is the Love Song Dying?*](#) places thousands of songs as individual dots and lets collective behaviour emerge, and we borrowed that idea for the Exposure Explorer. The aesthetic we were going for is exactly that: thousands of dots forming a shape.

The project was always going to be led by Adrien's feedback. Every row in the dataset is a photo he actually took, so whenever we found a pattern, the first question was: does this match what actually happened? Only he could answer that. Lea and Martina brought the technical and visualization side; Adrien kept the analysis grounded in reality.

We are all EPFL students, so we recognised the events, the associations, the campus rhythms in the data. It wasn't abstract. It was places we'd been and things we'd also lived through.

Every spike in the chart is a day Adrien remembers. The numbers already knew the story. We just had to find a way to show it.



A selection from the archive, where every photograph is also a row of numbers.

What we built and why

The scroll order tells a story. We start with time, asking when Adrien was shooting and how often, because you need to understand the person before the technical stuff means anything. Then gear and exposure: how did the equipment and technique change? The Sankey closes it: of 57k frames, how many survived?

Our original ambition was broader. Early on we wanted to show how Adrien's editing style had evolved in Lightroom, whether his use of shadows, clarity, or exposure correction had shifted over four years. The EDA showed it was much more complicated than expected. Lightroom sliders don't work in isolation: adjusting shadows automatically affects how exposure reads, and pulling one parameter shifts others. Any pattern we tried to surface was immediately entangled with everything else, and the variation was too personal and context-dependent to produce anything readable. The Sankey became the replacement: not how the photos were edited, but how ruthlessly they were culled.

Photo Activity

When does Adrien shoot, and does the rhythm repeat across years? Year-to-day drill-down. Weekly or monthly views were too coarse; only day-level granularity surfaces individual named sessions.

Gear Timeline

Which camera was active when, and how abruptly did transitions happen? Swimlane heatmap, one row per body, intensity normalised per camera so all bodies stay visible. A bar chart race was rejected: the Sony A7 IV at ~96% of shots would immediately swamp everything else.

Exposure Explorer

What exposure choices does Adrien make, and do they cluster by gear or scene? ~2,000 photographs as dots in a two-axis space. Spring physics animate transitions between axis pairs. Each combination gets an insight card written by Adrien.

Lens Race

How did the mix of lenses shift over time, and which focal lengths gained favour? Animated bar chart race of cumulative shots per interchangeable lens. Fixed-lens cameras excluded.

Workflow Funnel

Of 57k frames captured, how many survived to publication? Sankey from Shots → Selection → Edition → Published. The 160k-to-500 ratio is so extreme that D3's automatic layout produced unreadable overlaps, so we placed nodes manually.

Getting the scatter plot right

The chart shows 2,000 photographs, not all 56,882. Two reasons: spring-physics animation at 60fps becomes unusable past a few thousand points, and a random sample wouldn't work anyway, since the Sony A7 IV accounts for ~96% of all shots and would swamp everything else. Instead we use a stratified sample capped at 400 shots per camera per year, so every body stays visible.

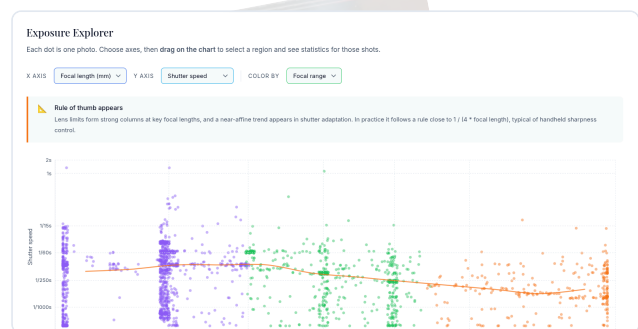
Log scales

On a linear axis, the gap between ISO 100 and ISO 200 looks identical to the gap between ISO 6400 and ISO 6500. But the first doubles the light sensitivity, a huge practical difference, while the second barely changes anything. The result: the range where most shots actually live gets squished into a corner, while the high end stretches out mostly empty. Switching to log scale means equal visual distance equals equal photographic difference, so the chart finally matched how the settings actually behave.

FIRST VERSION: YEAR AS DEFAULT, NO SELECTION



CURRENT: FOCAL RANGE DEFAULT, BRUSH + ZOOM



Colour and selection

Year, camera, and **focal range** (Wide / Normal / Tele) were all available as colour options from the start. What changed was understanding which one actually adds information: year tells you *when* a shot was taken, not *how*; camera groups by body but one body dominates. Focal range is independent of any axis and shows something the axes don't already carry. We also added a **binned median trend line** so the relationship is visible in the chart rather than just described in the card.

Selecting dots was missing entirely in the first version. We added a **D3 rectangular brush** with a live stats panel, and made releasing the brush zoom in: spring physics re-run and the cluster expands to fill the full canvas. A "← Zoom out" button brings everything back.

What a year actually looks like

Photo Activity

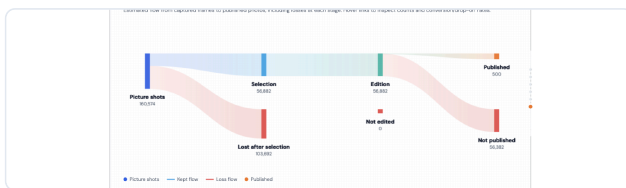
The year chart gave no sense of growth; the heatmap couldn't show which days belonged to the same session. The most useful addition was a named sessions bar chart below the heatmap, which tells you *what* each cluster of shooting was: Balélec, an association event, a sports day. The two views are bidirectionally linked so you can move between the pattern and the event behind it.

The year→heatmap transition went through many iterations before it felt right. A card flip and a height-collapse both looked broken. The final version: the remaining bars fade out, the selected bar zooms and expands until it fills the full canvas, then the heatmap content clips in from a rectangle that grows outward from the centre, so the bar becomes the heatmap container rather than being replaced by it.

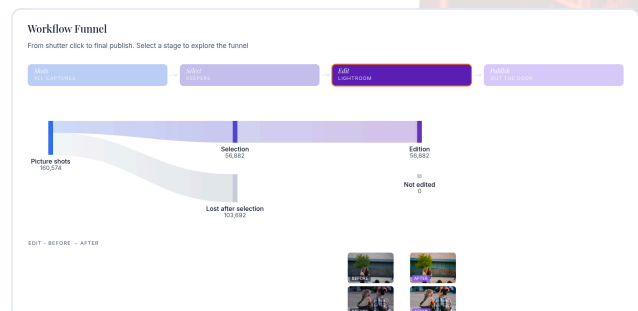
Workflow Funnel

The original Sankey showed all four stages at once. The Published node, 500 photos out of 160k, was easy to miss; the eye went straight to the big loss flow.

FIRST VERSION: EVERYTHING VISIBLE AT ONCE



CURRENT: STAGED REVEAL WITH PHOTO STRIP



We redesigned it as a **staged progressive reveal**, building left to right as the viewer advances. The first build re-animated all visible columns on every step, which we fixed by animating only the newly added stage. A photo strip shows what each stage looks like; at Edit, before/after pairs slide in side by side.

Behind the visualizations

The person whose data we were visualizing was also on the team. Every pattern had to hold up against someone who'd actually lived through the four years it describes.

Organising the archive

Before the project started, Adrien had already tagged every photo in his Lightroom library with a named collection: "Assos EPFL", "Balélec", "AGEPoly", and dozens more. All 56,882 photos carry a label. Without four years of quiet cataloguing, the sessions timeline in Photo Activity wouldn't be possible.

Shaping the narrative

An earlier draft opened with the Exposure Explorer. The pushback: a visitor landing on a scatter plot of ISO values has no idea whose photos these are. The site now opens with Photo Activity, time and volume first, and the technical charts come later.

The photo mosaic

124 photographs from the archive, curated and processed as WebP thumbnails, form the parallax grid that drifts as you scroll. The reveal effect, where photos materialise under the cursor across blank space, was a separate piece of the same idea. It's what visitors notice first.

Insight work for the Exposure Explorer

The chart was technically correct, but hard to read without photographic context. A cluster could be named: *handheld indoor events, high ISO, wide open*. All 12 axis combinations have insight cards covering the handheld sharpness rule, optical limits by lens family, and the aperture-priority shooting structure. Without someone who'd taken those photos, the plot is just dots.

What we got wrong first

The hardest design problem was the scatter plot axes. Photographers think in stops, each a doubling of light, so a linear scale misrepresents everything. The gap between $f/1.4$ and $f/2$ is a full stop; between $f/8$ and $f/9$ it's barely a third; on a linear axis they look identical. Switching to log scale made the clusters readable. Colour was the same problem: year tells you *when* a shot was taken, not *how*. Focal range turned out to be the most useful option, because it's independent of any axis and adds information the other two don't.

Interaction with the scatter plot went through a few versions. Dimming non-selected dots showed which points were chosen but gave no sense of how they clustered together. The zoom-into-selection approach works differently: spring physics re-run on the selection alone, the chosen points expanding to fill the canvas so their internal structure becomes readable.

The Sankey layout hit a hard constraint. D3's automatic algorithm places nodes by flow optimisation, which produced overlapping labels given the extreme 160k-to-500 funnel ratio. We ended up placing all nodes manually and using D3 only for bezier link rendering, which also let us control label positions without fighting the layout engine.

What stayed and why

Spring-physics dot animation. It makes clear that it's the same photos rearranging, not a new chart being drawn. It also handles the zoom animation for free.

Canvas + D3 SVG hybrid. Canvas handles 2,000 animated dots at 60fps; D3 SVG handles axes, brush, and trend line where correct hit-testing matters. Right call from the start.

Contextual insight cards. They exist because one person on the team could read clusters that the others could only see as dots. The text explains the photographic reason for each pattern, not just what the pattern is.

Scroll-driven single-page layout. Time → gear → technique → curation. An earlier draft opened with the Exposure Explorer; a scatter plot of ISO values means nothing without knowing whose data it is. The order stayed.

Who did what

WORK	LEA	ADRIEN	MARTINA
Exposure Explorer: axes, brush selection, zoom animation	●		
Exposure Explorer: insight cards (content & interpretation)		●	
Gear Timeline: swimlane heatmap, hover/drag, stats card	●		
Lens Race: animated bar chart race	●		
Photo Activity: banner, season colours, sessions chart, highlights			●
Year→heatmap transition animation	●	●	●
Workflow Funnel: original Sankey + before/after animation		●	
Workflow Funnel: staged reveal, photo strip, lightbox			●
Hero mosaic: curation, thumbnails, parallax (Adrien); cursor reveal effect (Lea)	●	●	
Site narrative, scroll order, colour palette		●	
Dataset (4 years of photography + Lightroom archive tagging)		●	
Data pipeline, year-range fix, camera naming	●		
Initial setup, EDA, project reports, process book	●	●	●

8 · REFLECTION

What we learned

Making five separate visualizations feel like one coherent thing was harder than any single technical problem. The order and framing matter as much as the charts, and we didn't get that right until late.

Working with personal data changed the dynamic. Every design decision had to hold up against someone who'd actually lived through the four years it describes. It also taught us that data cleaning isn't a pre-step: camera model normalisation, fixed-lens EXIF recovery, and stratified sampling are design decisions, as consequential as anything on the visual layer.